Control Method for Maintaining Dynamic Performance of a 6DOF Hybrid-reluctance-type Magnetic Levitator during Payload Mass Changes

Chang Hwan Kim[1], **Hyeong Min Yoon**[1], **Jae Woo Jung**[1], **Jun Young Yoon**[1,#]

1 Department of Mechanical Engineering, Yonsei University, Republic of Korea
Corresponding Author / E-mail: junyoung.yoon@yonsei.ac.kr

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6-degrees-of-freedom (6DOF) magnetic levitators have been widely used in industrial applications that demand precise positioning, e.g., a reticle stage in photolithography machines and 2-D/3-D scanning, since the levitation target can be supported by magnetic forces without any mechanical contact. This paper presents a control method designed to maintain dynamic performance of a 6DOF hybrid-reluctance-type magnetic levitator during payload mass changes. The levitator comprises a levitating platform that serves as a stage for the payload, and two iron-cored stator modules with permanent magnets (PMs) and electromagnets (EMs). In this setup, actively controlled magnetic fluxes generated by the EMs are super-imposed with the constant magnetic flux from the PMs, creating magnetic reluctance forces that levitate the stage. However, adding a payload on the stage changes the dynamics of the system, notably reducing the closed-loop bandwidth. It is critical to maintaining the closed-loop bandwidth of the system for precise positioning applications, since the ability to quickly respond to the reference position and effectively reject wide range of disturbances is closely related to the system bandwidth. Aiming to address such issue, this paper presents a control method, called mass compensator, which is essentially a modified first-order lead compensator. Since it works to cancel out the changed parts of its transfer function due to the payload mass without divergence in the time domain, it can recover the reduced closed-loop bandwidth of the levitation stage. In order to validate the proposed control method, measurements of the control loop frequency response and a comparison of the tracking performance for 2-D circular trajectories have been conducted under on-load and off-load conditions. Experimental results from the frequency responses show that the mass compensator can recover the crossover frequency of the levitation stage from 30 Hz to around 50 Hz while maintaining its phase margin. Furthermore, the increased RMS (root-mean-square) positioning error due to the payload mass is decreased by 98%, with the reference of 100 µm radius circular motion commanded at 5 Hz frequency. This indicates that by implementing the mass compensator in the controller, it achieves a level of error similar to that when no payload is present, effectively ensuring the dynamic performance of levitation stage remains nearly unchanged.

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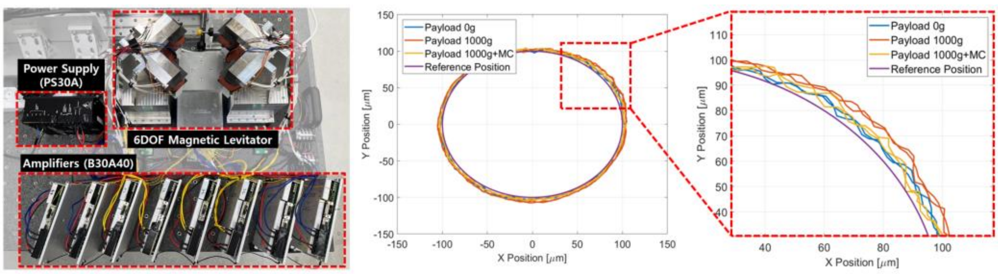

Fig. 1 Experiment setup (left) and tracking performance under 1 kg payload on the levitation stage (right). By implementing the mass compensator module, it achieves a level of error similar to the case of no payload